

Calculus, *distilled.*

*Every topic, every technique, every formula the course covers —
organised the way the exam asks for them. No appendix sheet is
given on the exam, so this guide is your reference.*

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§1 The whole course at a glance

The course covers 22 topics spread across 24 lectures, grouped by theme:

Foundations

- (Inverse) functions
- Limits of functions
- Asymptotes
- Implicit differentiation
- Linearization
- Differentials

Integration

- Substitution
- Integration by Parts
- Improper Integrals

Series

- Sequences
- Sum of series
- Convergence of a Series
- Power Series, Closed Form
- Interval of Convergence
- Taylor Series

Complex numbers

- Complex Numbers

Multivariate

- Multivariate Functions
- Multivariate Linearization
- Directional Derivatives
- Critical Points
- Double Integrals

Grade formula

$$\text{Final} = \frac{1}{3} \text{Midterm} + \frac{2}{3} \text{Endterm}$$

Midterm covers lectures 1–10 (mostly foundations + integration). Endterm covers everything.

§2 The formula vault

These are the facts you should know cold before walking into the exam — no formula sheet is provided.

Derivatives

- Power: $(x^n)' = nx^{n-1}$
- Exp / log: $(e^x)' = e^x$, $(\ln x)' = \frac{1}{x}$
- General exp: $(a^x)' = a^x \ln a$
- $(\sin x)' = \cos x$, $(\cos x)' = -\sin x$
- $(\tan x)' = \sec^2 x$, $(\cot x)' = -\csc^2 x$
- $(\sec x)' = \sec x \tan x$

- $(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$
- $(\arctan x)' = \frac{1}{1+x^2}$
- Product: $(fg)' = f'g + fg'$
- Quotient: $\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$
- Chain: $(f \circ g)' = f'(g) \cdot g'$
- Inverse: $(f^{-1})'(y) = \frac{1}{f'(f^{-1}(y))}$

Integrals (the ones you'll need on sight)

- $\int x^n dx = \frac{x^{n+1}}{n+1} + C, \quad n \neq -1$
- $\int \frac{1}{x} dx = \ln |x| + C$
- $\int e^x dx = e^x + C$
- $\int \sin x dx = -\cos x + C$
- $\int \sec^2 x dx = \tan x + C$
- $\int \frac{1}{1+x^2} dx = \arctan x + C$
- $\int \frac{1}{\sqrt{1-x^2}} dx = \arcsin x + C$
- $\int \tan x dx = -\ln |\cos x| + C$

Trig identities that save lives

- $\sin^2 x + \cos^2 x = 1$
- $1 + \tan^2 x = \sec^2 x$
- $\sin 2x = 2 \sin x \cos x$
- $\cos 2x = \cos^2 x - \sin^2 x$
- $\sin^2 x = \frac{1 - \cos 2x}{2}$
- $\cos^2 x = \frac{1 + \cos 2x}{2}$

Hyperbolics (appear occasionally)

$$\sinh x = \frac{e^x - e^{-x}}{2}, \quad \cosh x = \frac{e^x + e^{-x}}{2}, \quad (\sinh x)' = \cosh x, \quad (\cosh x)' = \sinh x$$

§3 Standard Maclaurin series

Memorise these. Almost every Taylor-series problem on the endterm reduces to *substituting*, *differentiating*, or *integrating* one of these six. Always know the radius of convergence too.

The six you must know

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \cdots, \quad R = 1, (-1, 1)$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \quad R = \infty$$

$$\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots, \quad R = \infty$$

$$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots, \quad R = \infty$$

$$\ln(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^n}{n} = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots, \quad R = 1, (-1, 1]$$

$$\arctan x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1} = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots, \quad R = 1, [-1, 1]$$

The three manipulation moves

If $f(x) = \sum a_n(x-c)^n$ on $(c-R, c+R)$, then on the same interval:

- **Substitute:** $f(g(x)) = \sum a_n(g(x)-c)^n$ — e.g. $e^{x^2} = \sum \frac{x^{2n}}{n!}$
- **Differentiate term-by-term:** $f'(x) = \sum n a_n(x-c)^{n-1}$
- **Integrate term-by-term:** $\int f = C + \sum \frac{a_n}{n+1}(x-c)^{n+1}$

Radius of convergence R stays the same; boundary behavior may change.

Worked example — compute $f^{(9)}(0)$ for $f(x) = e^{2x^3}$

Substitute $u = 2x^3$ into e^u : $f(x) = \sum \frac{(2x^3)^n}{n!} = \sum \frac{2^n x^{3n}}{n!}$.

The x^9 term comes from $n = 3$: coefficient is $\frac{2^3}{3!} = \frac{8}{6} = \frac{4}{3}$.

By Taylor's formula, coefficient of x^9 equals $\frac{f^{(9)}(0)}{9!}$, so $f^{(9)}(0) = \frac{4}{3} \cdot 9! = 483,840$.

§4 Convergence tests — the decision tree

Given $\sum a_n$, work down this list. The first test that applies gives you the answer.

Test	Use when	Conclusion
Divergence test	Always — check first	If $\lim a_n \neq 0$, diverges. If $= 0$, inconclusive.
Geometric	$a_n = ar^n$	Converges iff $ r < 1$; sum $= \frac{a}{1-r}$.
p-series	$a_n = 1/n^p$	Converges iff $p > 1$.
Integral test	f positive, decreasing, continuous	$\sum a_n$ and $\int_k^\infty f$ agree.
Comparison	$0 \leq a_n \leq b_n$	Bigger converges $\Rightarrow$ smaller converges.
Limit comparison	$\lim a_n/b_n = L \in (0, \infty)$	Both agree.
Ratio test	Involves $n!$, c^n	$L = \lim a_{n+1}/a_n $: < 1 conv., > 1 div.
Root test	Terms like $(b_n)^n$	$L = \lim \sqrt[n]{ a_n }$: < 1 conv., > 1 div.
Alternating series	$\sum (-1)^n b_n$, $b_n \geq 0$	Conv. if b_n eventually decreasing and $\rightarrow 0$.

Absolute vs conditional convergence

$\sum a_n$ is **absolutely convergent** if $\sum |a_n|$ converges; **conditionally convergent** if $\sum a_n$ converges but $\sum |a_n|$ diverges. Absolute convergence $\Rightarrow$ convergence, but not the reverse.

Alternating series error bound

$$|s - s_N| \leq b_{N+1}$$

The error from truncating at the N -th partial sum is bounded by the first omitted term's absolute value. Appears on *every* endterm for estimating integrals like $\int_0^{0.1} e^{-x^2} dx$ to a given tolerance.

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§5 Functions & inverses

Inverse functions

A function f is **invertible** on its domain iff it is **one-to-one**. Strictly monotonic functions are always one-to-one. To compute f^{-1} : write $y = f(x)$ and solve for x in terms of y .

Derivative of the inverse

$$(f^{-1})'(y) = \frac{1}{f'(f^{-1}(y))}$$

If $f(a) = b$, then $(f^{-1})'(b) = 1/f'(a)$ — no need for an explicit inverse formula.

§6 Limits & asymptotes

L'Hôpital's rule

Applies when $\lim f/g$ has form $0/0$ or ∞/∞ :

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

For other indeterminate forms: $0 \cdot \infty$ rewrite as $0/0$; $\infty - \infty$ combine fractions; $0^0, 1^\infty, \infty^0$ take logarithms.

Asymptotes

- **Vertical** $x = a$: $\lim_{x \rightarrow a^\pm} f = \pm\infty$. Usually from uncanceled zeros of denominator.
- **Horizontal** $y = L$: $\lim_{x \rightarrow \pm\infty} f = L$.
- **Slant** $y = mx + b$: $\lim_{x \rightarrow \infty} (f - (mx + b)) = 0$. Use long division for rationals.

Slant asymptote

For $f(x) = \frac{x^2+3x+1}{x-1}$, divide: $x^2 + 3x + 1 = (x-1)(x+4) + 5$, so $f(x) = x + 4 + \frac{5}{x-1}$. As $x \rightarrow \infty$, slant asymptote $y = x + 4$.

§7 Implicit differentiation

When y is defined implicitly by $F(x, y) = 0$, differentiate both sides w.r.t. x , treating y as a function of x (so $\frac{d}{dx}[y^n] = ny^{n-1}y'$), then solve for y' .

Find y' for $x^3 + y^3 = 6xy$

$$3x^2 + 3y^2y' = 6y + 6xy', \text{ so } y'(3y^2 - 6x) = 6y - 3x^2, \text{ giving } y' = \frac{2y - x^2}{y^2 - 2x}.$$

Logarithmic differentiation

Use when f has the form $g(x)^{h(x)}$: take $\ln$, differentiate implicitly, solve for f' .

$$f(x) = x^{\sin x}$$

$\ln f = \sin x \cdot \ln x$. Differentiate: $\frac{f'}{f} = \cos x \ln x + \frac{\sin x}{x}$, so $f'(x) = x^{\sin x} \left(\cos x \ln x + \frac{\sin x}{x} \right)$.

§8 Linearization & differentials

Linearization of f at a

$$L(x) = f(a) + f'(a)(x - a)$$

The best linear approximation to f near $x = a$. Choose a as a nearby “nice” value.

Differentials. If $y = f(x)$, $dy = f'(x) dx$. For error propagation, $\Delta y \approx f'(x) \Delta x$.

Linearize

217] Let $f(x) = x^{1/3}$, $a = 216$ (since $\sqrt[3]{216} = 6$).

$f'(x) = \frac{1}{3}x^{-2/3} = \frac{1}{3(x^{1/3})^2}$. At $a = 216$: $f'(216) = \frac{1}{3 \cdot 36} = \frac{1}{108}$.

$L(x) = 6 + \frac{1}{108}(x - 216)$, so $\sqrt[3]{217} \approx 6 + \frac{1}{108} \approx 6.00926$.

Trick: rewrite $x^{(n-1)/n}$ as $(x^{1/n})^{n-1}$ so the evaluation uses the n th root you already know.

When linearization = Taylor

The linearization of f at a equals the first-order Taylor polynomial at a . If a problem gives you a power series for g centered at c , the linearization of g at c is just the constant + linear term of the series.

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§9 Substitution (u-sub)

Reverse of the chain rule. Look for a piece $u = g(x)$ whose derivative $g'(x) dx$ also appears.

$$\int f(g(x)) g'(x) dx = \int f(u) du, \quad u = g(x)$$

Patterns. Inside of a composition $\Rightarrow$ try that as u . Denominator under a power, numerator = derivative $\Rightarrow u = \text{denominator}$. For definite integrals, **change limits**: new bounds $g(a)$ to $g(b)$.

Trig substitution

Form	Substitute	Identity used
$\sqrt{a^2 - x^2}$	$x = a \sin \theta$	$1 - \sin^2 = \cos^2$
$\sqrt{a^2 + x^2}$	$x = a \tan \theta$	$1 + \tan^2 = \sec^2$
$\sqrt{x^2 - a^2}$	$x = a \sec \theta$	$\sec^2 - 1 = \tan^2$

§10 Integration by parts

$$\int u dv = uv - \int v du$$

LIATE — **how to pick u** . Pick u from whichever appears earliest: **L**ogarithmic $\rightarrow$ **I**nverse trig $\rightarrow$ **A**lgebraic $\rightarrow$ **T**rigonometric $\rightarrow$ **E**xponential.

$$\int x \ln x \, dx$$

L before A, so $u = \ln x$, $dv = x \, dx$. Then $du = \frac{1}{x} dx$, $v = \frac{x^2}{2}$.

$$\int x \ln x \, dx = \frac{x^2}{2} \ln x - \int \frac{x^2}{2} \cdot \frac{1}{x} dx = \frac{x^2}{2} \ln x - \frac{x^2}{4} + C.$$

Tabular method

For $\int x^n e^{\alpha x} dx$ or $\int x^n \sin(\alpha x) dx$: differentiate the polynomial column, integrate the exp/trig column, alternate signs.

§11 Partial fractions

Use on rational $P(x)/Q(x)$ with $\deg P < \deg Q$ (if not, long-divide first).

Decomposition template:

- Distinct linear $(x - a)$: $\frac{A}{x-a}$.
- Repeated $(x - a)^k$: $\sum_{i=1}^k \frac{A_i}{(x-a)^i}$.
- Irreducible quadratic $(x^2 + bx + c)$: $\frac{Ax+B}{x^2+bx+c}$.

$$\int \frac{x+4}{x^2-5x+6} dx$$

Factor: $x^2 - 5x + 6 = (x - 2)(x - 3)$. Write $\frac{x+4}{(x-2)(x-3)} = \frac{A}{x-2} + \frac{B}{x-3}$.

Cover-up: $A = \frac{2+4}{2-3} = -6$, $B = \frac{3+4}{3-2} = 7$.

Integral = $-6 \ln|x-2| + 7 \ln|x-3| + C$.

§12 Trigonometric integrals

Patterns for $\int \sin^m x \cos^n x \, dx$:

- m or n odd: peel off one factor, convert rest via $\sin^2 + \cos^2 = 1$, then u-sub.
- Both even: use power-reduction $\sin^2 = \frac{1-\cos 2x}{2}$, $\cos^2 = \frac{1+\cos 2x}{2}$.

§13 Improper integrals

An integral is *improper* if the interval is unbounded or the integrand has a singularity. Always evaluated as a limit.

Types

Type I (infinite interval): $\int_a^\infty f \, dx = \lim_{t \rightarrow \infty} \int_a^t f \, dx$.

Type II (vertical asymptote at c): split at c , take one-sided limits.

Converges if the limit is finite; otherwise diverges.

The p -integrals — know these cold

$$\int_1^\infty \frac{1}{x^p} dx \text{ converges iff } p > 1; \quad \int_0^1 \frac{1}{x^p} dx \text{ converges iff } p < 1.$$

Exam variant: $\int_e^\infty \frac{1}{x(\ln x)^p} dx$ converges iff $p > 1$ (substitute $u = \ln x$).

$$\int_0^\infty e^{-\sqrt{x}} dx$$

Substitute $u = \sqrt{x}$, $x = u^2$, $dx = 2u du$. Limits: $0 \rightarrow 0$, $\infty \rightarrow \infty$.

$$\int_0^\infty 2ue^{-u} du. \text{ IBP: } = [-2ue^{-u}]_0^\infty + \int_0^\infty 2e^{-u} du = 0 + 2 = 2.$$

Comparison test for integrals

If $0 \leq f \leq g$ on $[a, \infty)$ and $\int g$ converges, then $\int f$ converges. Useful when you can bound by a p -integral.

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§14 Sequences

$\{a_n\}$ **converges** to L if $\lim a_n = L$. Tools: L'Hôpital on $a(x)$; squeeze theorem; monotone + bounded $\Rightarrow$ converges.

Key limits: $\lim \frac{\ln n}{n^p} = 0$; $\lim r^n = 0$ if $|r| < 1$; $\lim \sqrt[n]{n} = 1$; $\lim(1 + \frac{x}{n})^n = e^x$.

Recursive sequences & Newton's method

If $a_{n+1} = f(a_n) \rightarrow L$ with f continuous, then $L = f(L)$ (fixed point). **Newton:** $a_{n+1} = a_n - f(a_n)/f'(a_n)$.

§15 Series & summing

$\sum a_n$ converges to s if partial sums $s_N \rightarrow s$.

Closed-form sums to know

- Geometric: $\sum_{n=0}^\infty ar^n = \frac{a}{1-r}$, $|r| < 1$.
- Finite geometric: $\sum_{n=0}^N ar^n = \frac{a(1-r^{N+1})}{1-r}$.
- Telescoping: $\sum_{n=1}^N (b_n - b_{n+1}) = b_1 - b_{N+1}$.
- From e : $\sum_{n=0}^\infty \frac{1}{n!} = e$.

Evaluate $\sum_{n=0}^{\infty} \frac{n!+2^n}{n!2^n}$
 Split: $\sum \frac{n!+2^n}{n!2^n} = \sum \frac{1}{2^n} + \sum \frac{1}{n!} = \frac{1}{1-1/2} + e = 2 + e$.

Partial fractions on series

To sum $\sum \frac{1}{n(n+1)}$: write $\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$, then telescope.

§16 Convergence tests (deep dive)

See the decision tree in §4. Heuristic for picking a test:

- Terms don't go to zero $\Rightarrow$ divergence test. Done.
- Contains $n!$, c^n , $n^n \Rightarrow$ ratio test (root if whole term is an n th power).
- Rational in n / involves $\ln n$, $\sqrt{n} \Rightarrow$ limit comparison with a p -series.
- Contains $(-1)^n$ with terms $\rightarrow 0$ monotonically $\Rightarrow$ alternating series test.
- Looks like $f(n)$, f decreasing, $\int f$ easy $\Rightarrow$ integral test.

Classify $\sum (-1)^n n^2 e^{-n}$

Absolute version: $\sum n^2 e^{-n}$. Ratio: $\frac{(n+1)^2}{n^2} \cdot e^{-1} \rightarrow \frac{1}{e} < 1$. Converges absolutely.

Classify $\sum \frac{1+\sin n}{n^2}$

All terms ≥ 0 . Bound: $0 \leq \frac{1+\sin n}{n^2} \leq \frac{2}{n^2}$. $\sum \frac{2}{n^2}$ converges ($p = 2$), so does ours by comparison. Absolutely convergent.

§17 Power series & interval of convergence

Definition

A **power series** centered at c is $\sum a_n(x-c)^n$. Converges absolutely on $(c-R, c+R)$ where R is the **radius of convergence**. At $x = c \pm R$, behavior must be checked separately.

Finding R : $R = \lim |a_n/a_{n+1}|$, or apply the ratio test to the whole power series.

Four-step procedure (memorise)

1. Ratio (or root) test to find where converges absolutely. Gives R and $(c-R, c+R)$.
2. Plug in $x = c - R$; test using any test from the table.
3. Plug in $x = c + R$; test.
4. Report the interval with correct brackets.

$$\sum_{n=1}^{\infty} \frac{\ln n}{n \cdot 2^n} (x-2)^n$$

Step 1. Ratio: $\lim \left| \frac{\ln(n+1) n 2^n}{(n+1) 2^{n+1} \ln n} \right| \cdot |x-2| = \frac{|x-2|}{2}$. Converges for $|x-2| < 2$, so $x \in (0, 4)$, $R = 2$.

Step 2. At $x = 0$: $\sum (-1)^n \frac{\ln n}{n}$. Alternating, $\frac{\ln n}{n} \rightarrow 0$ (L'H), decreasing for $n > e$. Converges by AST. Include.

Step 3. At $x = 4$: $\sum \frac{\ln n}{n}$. Since $\frac{\ln n}{n} \geq \frac{1}{n}$ for $n \geq 3$, diverges (harmonic). Exclude.

Answer: $[0, 4)$.

Closed-form from a power series

Recognise the series as a manipulation of a standard Maclaurin series.

- $\sum x^n = \frac{1}{1-x} \rightarrow$ differentiate: $\sum nx^{n-1} = \frac{1}{(1-x)^2}$.
- $\sum \frac{x^n}{n} = -\ln(1-x)$ (integrate geometric).
- $\sum \frac{x^n}{n+1} = -\frac{\ln(1-x)}{x}$.

§18 Taylor series

Taylor series of f centered at a

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n$$

When $a = 0$, it's the **Maclaurin series**.

Two strategies. (1) Direct: compute $f^{(n)}(a)$ — only when a pattern emerges. (2) Manipulation: start from a standard series, substitute / differentiate / integrate / multiply. Usually faster.

First 3 non-zero terms of $\int x^3 e^{-3x^3} dx$

$e^u = 1 + u + \frac{u^2}{2} + \dots$. Sub $u = -3x^3$: $e^{-3x^3} = 1 - 3x^3 + \frac{9x^6}{2} - \dots$

Multiply by x^3 : $x^3 - 3x^6 + \frac{9x^9}{2} - \dots$

Integrate: $C + \frac{x^4}{4} - \frac{3x^7}{7} + \frac{9x^{10}}{20} - \dots$

First three non-zero terms: $\frac{x^4}{4}, -\frac{3x^7}{7}, \frac{9x^{10}}{20}$.

The $f^{(k)}(0)$ trick

Build the Maclaurin series by substitution, read off the coefficient of x^k , then

$$f^{(k)}(0) = k! \cdot (\text{coefficient of } x^k).$$

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§19 Complex numbers

Three forms of the same number

Rectangular: $z = a + bi$, $a = \Re(z)$, $b = \Im(z)$.

Polar: $z = r(\cos \theta + i \sin \theta)$, modulus $r = |z| = \sqrt{a^2 + b^2}$, argument $\theta = \arg(z)$.

Polar exponential: $z = re^{i\theta}$.

Conversion: $a = r \cos \theta$, $b = r \sin \theta$, $\theta = \arctan(b/a)$ (adjust for quadrant!).

Arithmetic

- Sum/difference: component-wise, rectangular form best.
- Product: $(a + bi)(c + di) = (ac - bd) + (ad + bc)i$, or polar: $|zw| = |z||w|$, $\arg(zw) = \arg z + \arg w$.
- Division: multiply by conjugate $\bar{w} = c - di$.
- Conjugate: $\overline{a + bi} = a - bi$, $|z|^2 = z\bar{z}$.

De Moivre's formula

$$(re^{i\theta})^n = r^n e^{in\theta} = r^n (\cos n\theta + i \sin n\theta)$$

Euler's identity (prove via power series)

$$e^{i\theta} = \cos \theta + i \sin \theta$$

Plug $i\theta$ into the Maclaurin series for e^x , separate even/odd terms using $i^{2n} = (-1)^n$, $i^{2n+1} = (-1)^n i$, recognise $\cos \theta$ and $\sin \theta$.

n th roots & complex equations

n th roots of $w = Re^{i\varphi}$

$$z_k = R^{1/n} \exp\left(i \frac{\varphi + 2\pi k}{n}\right), \quad k = 0, 1, \dots, n-1.$$

The n roots lie on a circle of radius $R^{1/n}$, equally spaced by $2\pi/n$ — a regular n -gon.

All solutions to $(z - 2)^3 = -8$

$-8 = 8e^{i\pi}$, so $z - 2 = 2e^{i(\pi + 2\pi k)/3}$, $k = 0, 1, 2$.

- $k = 0$: $z - 2 = 2e^{i\pi/3} = 1 + i\sqrt{3}$, so $z = 3 + i\sqrt{3}$.
- $k = 1$: $z - 2 = 2e^{i\pi} = -2$, so $z = 0$.
- $k = 2$: $z - 2 = 2e^{i5\pi/3} = 1 - i\sqrt{3}$, so $z = 3 - i\sqrt{3}$.

Quartics that reduce

Equations like $z^4 - iz^2 + 3 = 1$ are quadratic in z^2 — solve for z^2 first, then take square roots of each.

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§20 Partial derivatives

For $f(x, y)$: $f_x = \partial f / \partial x$ (treat y as constant), $f_y = \partial f / \partial y$ (treat x as constant). Second partials $f_{xx}, f_{yy}, f_{xy}, f_{yx}$. Clairaut: $f_{xy} = f_{yx}$ for nice f .

$$f(x, y) = y^x$$

Rewrite $y^x = e^{x \ln y}$. Then $f_x = y^x \ln y$, $f_y = xy^{x-1}$.

§21 Gradient & directional derivatives

Gradient

$$\nabla f(x, y) = \langle f_x, f_y \rangle$$

Vector pointing in the direction of fastest increase.

Directional derivative

For a **unit** vector $\mathbf{u}$: $D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u}$.

If $\mathbf{v}$ not unit, normalise: $\mathbf{u} = \mathbf{v} / \|\mathbf{v}\|$.

Geometric meaning. $D_{\mathbf{u}}f$ = rate of change of f in direction $\mathbf{u}$. Max rate at a point is $\|\nabla f\|$, in direction ∇f . $\nabla f \perp$ level curve at that point.

§22 Multivariate linearization

Linearization of $f(x, y)$ at (a, b)

$$L(x, y) = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$$

Equation of the *tangent plane* to $z = f(x, y)$ at $(a, b, f(a, b))$.

Tangent plane in 3D: $z - c = f_x(a, b)(x - a) + f_y(a, b)(y - b)$.

§23 Critical points & extrema

Critical point: $\nabla f = \mathbf{0}$, i.e. $f_x = f_y = 0$ simultaneously.

Second-derivative (Hessian) test

$D = f_{xx}f_{yy} - (f_{xy})^2$ at a critical point (a, b) :

- $D > 0, f_{xx} > 0$: **local minimum**.
- $D > 0, f_{xx} < 0$: **local maximum**.
- $D < 0$: **saddle point**.
- $D = 0$: inconclusive.

Absolute extrema on a closed region D

1. Find critical points in the **interior**; evaluate f at each.
2. Find extrema on the **boundary**: parametrise each piece, reduce to 1D.
3. Check the **corners**.
4. Compare all values.

$h(x, y) = 1 - 34x - 12y + 8xy + 8x^2$ on triangle $(1, 1), (2, 1), (2, 2)$

Interior: $h_x = -34 + 8y + 16x = 0$, $h_y = -12 + 8x = 0$. Second gives $x = 3/2$; first gives $y = 5/4$. Check whether $(3/2, 5/4)$ is inside the triangle $\{1 \leq x \leq 2, 1 \leq y \leq x\}$ and evaluate if so.

Boundaries: parametrise the three edges, find 1D critical points.

Vertices: evaluate $h(1, 1), h(2, 1), h(2, 2)$ directly.

Compare; pick largest and smallest.

§24 Double integrals

Over a rectangle $R = [a, b] \times [c, d]$ (Fubini)

$$\iint_R f \, dA = \int_a^b \int_c^d f(x, y) \, dy \, dx = \int_c^d \int_a^b f \, dx \, dy$$

Over a general region D

Type I ($a \leq x \leq b, g_1(x) \leq y \leq g_2(x)$):

$$\iint_D f \, dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f \, dy \, dx$$

Type II ($c \leq y \leq d, h_1(y) \leq x \leq h_2(y)$):

$$\iint_D f \, dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f \, dx \, dy$$

Switching order

If the inner integral is impossible (e.g. $\int e^{-y^2} dy$ has no elementary form), swap the order. Sketch the region, re-describe using the other variable as outer.

Polar coordinates

$$\iint_D f(x, y) dA = \iint_{D'} f(r \cos \theta, r \sin \theta) r dr d\theta$$

Don't forget the extra r in the Jacobian.

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§25 Exam strategy**Time budget (2 hours typically)**

Short answers first (30s–2min each). Open questions: budget by points — 10 pts $\approx$ 12 min. Leave 10 min at the end for checking.

General tactics

- **Always show work.** Rubrics reward argumentation — stating “converges by ratio test” without computing the limit loses points.
- **Argue convergence claims fully.** Alternating series conclusion needs: (1) terms alternate, (2) eventually decrease, (3) go to zero. Don't skip (2).
- **Start Taylor expansions from a known series** — faster and harder to get wrong than computing $f^{(n)}$ directly.
- **Simplify before substituting** in partial-fractions / complex problems.
- **Sketch** when the problem involves regions, complex roots, or asymptotes.
- **Check endpoints** on power series and improper integrals.

Common point-loss patterns

- Stating interval of convergence as $(c - R, c + R)$ without checking endpoints.
- Forgetting the Jacobian r in polar double integrals.
- Forgetting the $+C$ on indefinite integrals.
- Using a non-unit vector directly in $D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u}$ without normalising.
- Skipping the monotonicity check in the alternating series test.
- Dropping a factor when differentiating / integrating a series term-by-term.
- Wrong center: $\sum c_n(x^2 - 1)^n$ is a series in $(x^2 - 1)$, not in x .

The night before — checklist

- ☐ Six standard Maclaurin series + radii from memory.
- ☐ Nine convergence tests and when each applies.
- ☐ p -series and p -integral thresholds (and the $\ln$ variant).
- ☐ Solve $z^n = w$ for all n roots, place on complex plane.
- ☐ Classify critical points with Hessian test.
- ☐ Boundary optimisation on closed regions (triangle, disk).
- ☐ Four trig substitution identities.
- ☐ Swap order of a double integral.
- ☐ Convert double integral to polar with Jacobian.
- ☐ Derive Euler's formula from power series.

If you can say yes to all of these — you're ready.

Go through one past exam under timed conditions the morning before. Don't learn new material the night before. Sleep.

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Study guide compiled from CSE1200 lecture materials, Stewart's Calculus: Early Transcendentals 9e, and past exams. Good luck.